

AF Identification From Time–Frequency Analysis of ECG Signal Using Deep Neural Networks

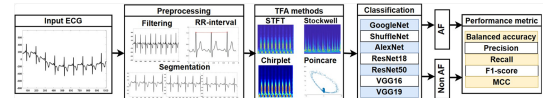
Thivya Anbalagan¹  and Malaya Kumar Nath¹ 

Department of Electronics, Communication Engineering, National Institute of Technology Puducherry, Karaikal 609609, India

*Senior Member, IEEE

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Abstract—Atrial fibrillation (AF) is a life threatening cardiac abnormality having high prevalence and risk with increased rate of stroke and systemic embolism, if oral anticoagulation is not recommended. Later, this leads to morbidity and mortality. Detection of AF is challenging from the electrocardiogram (ECG) recordings, due to its complex characteristics. Manual observation of ECG is tedious, time consuming, and error prone. This manuscript proposed a novel approach for identifying AF in the presence of noise and other beats by using deep neural networks (DNN) on the 2-D patterns obtained by various time–frequency analysis methods, such as short time Fourier transform (STFT), Chirplet-transform, Stockwell-transform, and Poincare plot from 1-D preprocessed ECG recordings. The above discussed methods identify the variations due to AF in ECG. Initially, the patterns are used by the pretrained DNN models for classification. ResNet18 attained the highest accuracy of 90.56% for the patterns of ECG obtained by Chirplet-transform on PAF prediction challenge database, whereas Chirplet patterns used by ResNet50 achieved an accuracy of 89.72%. Based on accuracy and the number of parameters in DNN, an ensemble network is designed for improving AF classification. Ensembling of ShuffleNet and AlexNet applied over the patterns obtained by Stockwell transform achieved the highest accuracy of 93.70%. This approach is further experimented on PhysioNet CinC 2017 challenge database, consisting of four classes (such as AF, normal, other rhythms, and noise).



Index Terms—Sensor applications, atrial fibrillation (AF), chirplet, ensemble deep neural network (DNN), poincare, short time Fourier transform (STFT), stockwell.

I. INTRODUCTION

Atrial fibrillation (AF) is an irregular heart rhythm affecting the upper chambers of heart (atria), which is one of the leading cause of mortality globally with an annual healthcare cost of 26 billion USD in the United States every year [1]. During normal heart rhythm, atria contracts and pumps blood into the lower chambers (i.e., ventricles) in a coordinated and organized manner. In AF, atria quiver instead of contracting effectively. This leads to chaotic and disorganized electrical signals (having irregular and rapid heartbeats) [2]. Ineffective contractions raise the chance of blood clot formation in the atria, which can cause a stroke, if it goes to the brain. Age (over 60), hyperthyroidism, obesity, untreated sleep apnea, coronary artery disease, heart valve disorders, and heart failure are few causes for increasing the risk of AF. The common symptoms of AF include heart palpitations (feeling rapid, irregular, or fluttering heartbeats), fatigue (unexplained tiredness or lack of energy), dizziness or lightheadedness (feeling faint or dizzy), and shortness of breath (difficulty breathing or a feeling of breathlessness, especially during physical activity) [3]. Some people may experience chest pain or discomfort, but this is less common. Some individuals with AF may not experience any symptoms at all, and the condition may be detected during routine medical examinations or investigations for other health issues. Medication, cardio-version, ablation, lifestyle changes, and blood thinning are the treatment plan for their specific conditions of AF. Managing the above

discussed conditions and adopting a heart-healthy lifestyle are crucial in preventing and managing AF.

Healthcare professionals can diagnose AF through tests, such as electrocardiogram (ECG or EKG) and recommended appropriate treatment options, which may include medications, lifestyle changes or medical procedures [4]. During AF, variations are noticed in ECG signals, such as absence of a regular P-wave pattern, irregular RR-intervals, narrow QRS complex, and an irregular ventricular response. Irregularity in RR-interval represents collapse and irregular atrial activity [5]. P-wave represents atrial depolarization. In AF condition, fibrillatory or erratic waves are noticed in ECG, indicating disorganized atrial electrical activity. AF condition can lead to a rapid and irregular ventricular rate. This is often observed as a high heart rate (varies among persons). QRS complexes (representing ventricular depolarization) are typically narrow, indicating that the electrical impulse is originating in the atria and conducting through the normal pathway.

A. Literature Survey

This part discusses about various approaches, procedures, and performance metric used by the researchers for identification and classification of AF from ECG signals along with their limitations. Conventional signal processing (i.e., spectrogram, P-wave detection, entropy features, and heart rate variability) and deep learning (DL) techniques are mainly used for AF identification from the ECG recordings [6]. Conventional approaches are less accurate compared to DL techniques, but the later one is more noise-sensitive and requires more computational time and training data. Works carried out on this area in last couple of decades are surveyed and the effective methods for AF detection and classification are summarized in

Corresponding authors: Thivya Anbalagan; Malaya Kumar Nath (e-mail: thivyaanbu@gmail.com; malaya.nath@gmail.com).

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Table 1. Recently Developed Techniques for Detecting AF From ECG Recordings

S. No	Authors & Year	Technique	Dataset	Performance: Accuracy (%)	Limitations
1	Parmar et al., (2023) [8]	DNN, VLSI architecture	PhysioNet MIT-BIH AF	92.37	Less data
2	Akiliu et al., (2023) [9]	AliveCor heart monitor	Real-time	-	Cost-effective
3	Voinas et al., (2023) [10]	VAE	CACHE-T-CADB	92.83	Presence of noise was not analyzed
4	Zhang et al., (2023) [11]	* CNN-LSTM, * RF, * PDFA	cECG from UnoVIS clin2009	87.40	* Highly imbalanced data, * Performance affected by noise, * Artifacts are sensitive to P- and Q-wave
5	Zhang et al., (2023) [12]	* DWT for feature enhancement, * Data imbalance processing, * MP-DLNet-F	* CinC 2017, * AFDB, * CPSC 2018	92.70	* Poor generalization, * High imbalance data
6	Huerta et al., (2024) [13]	Phase space plot	Real-time data	94	Data mislabeling
7	Mosheh et al., (2023) [14]	RawECGNet	* UVAF, * Rambam Hospital, * Saitama Hospital	F1-score - 0.94	Less performance
8	Zhu et al., (2022) [15]	Hybrid (ML + statistical) decision model	* Watch PPG, * Watch ACC	57.18	* Algorithm fails in presence of other rhythms and AFL, * Subjects having PAF were not identified
9	Yebra et al., (2022) [16]	π component analysis for QT delineation	* Ambulatory ECG, * Simulated ECG	-	Model performance for noise > 20 μ V were not investigated

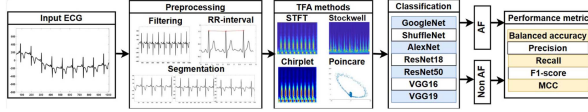


Fig. 1. Block diagram for the proposed AF identification using DNN over time–frequency representation of ECG recordings.

Table 1. Mainly researchers used MIT-BIH AF database for identifying AF, but it lacks of AF annotations for validating the model [7]. In most cases, models were developed with a large number of layers and high computational parameters, resulting in poor classification performance due to increased false positives (FP) and a lack of annotated ECG signals. Recently developed DL methods achieved the highest accuracy of 92.83%, and fails in accurate identification of AF in presence of other rhythms, noise, and other arrhythmia conditions. Data recorded from local hospitals has privacy and security concerns. Labeling these data are time consuming, expensive, and error prone. Some models trained with limited annotated data and achieved poor generalization for unseen data. The rest of this letter is organized as follows. The proposed method for classification and detection of AF has been represented in Section II. The contributions made in this work are as follows. 1) Pre-processing by second order Butterworth filter followed by Stockwell transform (ST) for detection of R-peak and RR-intervals. 2) Identifying the temporal variations due to AF by graphical representation using time–frequency analysis (TFA), such as, Stockwell, Chirplet, short time Fourier transform (STFT), and Poincare. 3) Classification of AF using deep neural network (DNN) followed by performance evaluation metric [such as balanced accuracy, precision, recall, F1-score, and Matthews correlation coefficient (MCC)]. Results and discussions are summarized in Section III along with the potential strength of the proposed approach compared to the state-of-the-art. Finally, Section IV concludes this letter.

II. PROPOSED METHODOLOGY

Fig. 1 depicts the systematic procedures for the proposed AF detection technique, which includes preprocessing, 2-D representation of ECG signal, classification, and performance evaluation. The preprocessing step consists of mean subtraction followed by a second order Butterworth filter for removing low frequency noise, i.e., baseline wander and power-line interference. Butterworth filters with

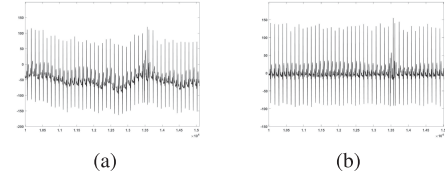


Fig. 2. ECG recordings. (a) Original signal and (b) filtered signal.

Table 2. Representation for TFA Techniques

S.No	Methods	Mathematical representation	TFA for normal ECG	TFA for AF
1	STFT	$X(t, f) = \int_{-\infty}^{\infty} x(\tau) \omega(\tau - t) e^{-j2\pi f\tau} d\tau$		
2	Stockwell	$S_s(t, f) = \int_{-\infty}^{\infty} x(\tau) g(\tau - t, f) e^{-j2\pi f\tau} d\tau$		
3	Chirplet	$C_s(a, b, \omega) = \int_{-\infty}^{\infty} x(t) \psi_{a,b,\omega}^*(t) dt$		
4	Poincare	$d_1 = \frac{RR(n) - RR(n-1)}{\sqrt{RR(n)}}$		
Where, $X(t, f)$ - STFT of the signal, $x(\tau)$ - Original signal, $\omega(\tau - t)$ - Window function		$S_s(t, f)$ - ST of the signal t - Time f - Frequency a - Scale	b - Time-shift ω - Frequency parameters $\psi_{a,b,\omega}(t)$ - Chirplet waveform $RR(n)$ - n^{th} RR-interval	

lower order have less roll-off compared to higher orders. But, the later introduces phase shift. As a result, a tradeoff exists between order and filtering performance, as well as the phase shift. Graphical representation of original- and filtered-signal are shown in Fig. 2. This filtered signal is subjected to ST for determining RR-intervals followed by time–frequency representations. ST depends on STFT concept and an effective tool for joint TFA resolution, which provides frequency sustaining direct link with the Fourier spectrum with a moving and scalable Gaussian window as its basis. Mathematically, it is defined as

$$S(\tau, f) = \frac{|f|}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x(t) e^{-\frac{(t-\tau)^2}{2}} e^{-j2\pi f t} dt \quad (1)$$

where $x(t)$, f , and τ represent signal of interest, its' frequency, and window's function control parameter for determining position of Gaussian window on time axis. The preprocessed ECG signals are subjected to STFT, Chirplet, ST, and Poincare plot for representing signal characteristics both in time and frequency domain simultaneously. These methods represent the variations in ECG recordings by specific patterns in form of 2-D images. Mathematical representation for TFA along with their graphical interpretations are illustrated in Table 2. STFT observes the variations in a signal using overlapping segments [17]. ST improves over STFT by incorporating a window function varying with frequency, which provides better time–frequency localization compared to spectrogram [18]. Chirplet transform analyses both instantaneous frequency and frequency modulation of a signal [11]. Poincare identifies the dynamic system by computing the relationship between current RR- and previous RR-intervals. Features extracted from TFA techniques are used by DNN (such as AlexNet, ShuffleNet, VGG16, VGG19, ResNet18, NasNet-mobile, ResNet50, and GoogleNet) for AF identification.

In these networks, fully connected-, Softmax-, and classification-layers are modified based on the number of classes. The best-performing networks, based on the performance metric values (such as balanced accuracy, precision, recall, F1-score, and MCC) and having least number of parameters are ensembled for classification [19]. GoogleNet is ensembled with ShuffleNet, ResNet18, and AlexNet while AlexNet is ensembled with ShuffleNet and ResNet18. Ensembling process combines the input layer of different networks for transmitting the same information. To obtain the classified output, the fully connected layers are modified based on the number of input

classes and the features are concatenated followed by fully connected-, Softmax-, and classification-layer.

A. Database

The proposed AF detection method is experimented on PAF prediction challenge database having four record sets (consists of two 30-min- and two 5-min-records) obtained from 48 different patients [7]. The abbreviations “p” and “n” represent PAF and normal, respectively. The number of normal and AF ECG recordings are increased to 1200 and 600, respectively, by splitting each original ECG signals into a segment of 3000. This work is further experimented on PhysioNet CinC 2017 challenge database having 9–61 s of recordings obtained from 8528 subjects, using handheld single-lead ECG recorder, provided by AliveCor. It has 4-classes, such as AF, N, other rhythms, and noisy with 504, 3695, 1655, and 123 number of recordings, respectively.

B. Performance Evaluation

The proposed model’s ability for detecting the diagnostic information is assessed from confusion matrix by evaluating balanced accuracy, precision, recall, F1-score, and MCC [6]. Balanced accuracy represents averaging of sensitivity and specificity. In binary classification, F1-score is used to quantify a model’s precision and recall. Precision measures the quality of positive predictions, defined by the ratio of true positives (TP) to the sum of “TP” and FP. Recall measures correctly classified positive samples, which is defined as the ratio of “TP” to the sum of “TP” and false negatives. MCC is a better measure for imbalanced class to identify the difference between predicted- and actual-values.

III. RESULTS AND DISCUSSION

The proposed model has been experimented by the system with Intel Core i7 7700 CPU @ 3.60 GHz with memory 24 GB, which includes NVIDIA GeForce RTX 3070 graphics of 8 GB and 405 MHz, manufactured by ZOTAC International Ltd.

AF classification performance using pretrained DNN models on image features obtained by Poincare, Chirplet, STFT, and ST have been experimented on ECG signals available in PAF prediction challenge- and PhysioNet CinC 2017 challenge-databases. Each ECG signal in the database are segmented to a length of 3000 samples to generate 1800 signals (1200 normal and 600 AF) to obtain the 2-D visual representations by TFA approaches. In total, 80% of the images from the two classes are used for training by the DNN models, whereas the rest 20% are considered for testing. The model used SGDM optimizer with a learning rate of 0.001, for different epochs, validation frequencies, and mini batch sizes. Epoch and learning rate are modified iteratively to obtain the optimized classification performance, out of which the best performing models along with the 2-D features are ensembled for classification.

Table 3 represents the performance accuracy of various DL models (such as AlexNet, ResNet18, ShuffleNet, GoogleNet, VGG16, VGG19, and ResNet50 networks) with their depth and number of parameters for different TFA methods. Across various models, AlexNet, ResNet18, and ResNet50 performed better for the features obtained by Chirplet transform and recorded an accuracy of 86.37%, 90.56%, and 89.72%, respectively. For STFT features, ShuffleNet and GoogleNet recorded an accuracy of 87.78% and 89.44%, respectively. VGG16 and VGG19 attained the highest accuracy of 80.77% and 78.78% for Poincare- and Stockwell-transformed-images, respectively.

TABLE 3. Classification Performance by DNN for TFA (2-D Images) From PAF- and CinC 2017-Database

S. No	Networks	Depth	Parameters (Millions)	TFA methods	Accuracy (%) for PAF data	Accuracy (%) for CinC 2017 data
1	AlexNet	8	61.0	Poincare	77.78	77.75
				Stockwell	83.33	80.00
				STFT	80.83	77.50
				Chirplet	86.37	80.00
2	ResNet18	18	11.7	Poincare	77.78	77.50
				Stockwell	87.78	90.00
				STFT	86.67	87.50
				Chirplet	90.56	82.50
3	ShuffleNet	50	1.4	Poincare	82.22	82.50
				Stockwell	85.83	87.50
				STFT	89.44	85.00
				Chirplet	87.78	85.00
4	GoogleNet	22	7.0	Poincare	80.77	77.50
				Stockwell	75.00	90.00
				STFT	67.80	82.50
				Chirplet	63.33	82.50
5	VGG16	16	138	Poincare	76.92	72.50
				Stockwell	78.76	85.00
				STFT	67.80	82.50
				Chirplet	66.67	75.00
6	VGG19	19	144	Poincare	76.92	75.00
				Stockwell	78.76	82.50
				STFT	67.80	77.50
				Chirplet	66.67	82.50
7	ResNet 50	50	25.6	Poincare	83.61	75.00
				Stockwell	86.67	87.50
				STFT	76.84	82.50
				Chirplet	89.72	85.00

TABLE 4. Classification Result of Ensembled DNN for 2D TFA Representation of 1D ECG Signal

S. No	Ensembled networks	2D image	Accuracy (%) for PAF data	Accuracy (%) for CinC 2017 data
1	GoogleNet + ShuffleNet	Poincare	81.58	82.50
		Stockwell	88.33	85.00
		STFT	87.22	80.00
		Chirplet	87.50	82.50
2	ShuffleNet + AlexNet	Poincare	84.62	80.00
		Stockwell	93.70	87.50
		STFT	78.53	85.00
		Chirplet	88.89	90.00
3	AlexNet + ResNet18	Poincare	84.72	75.00
		Stockwell	85.00	87.50
		STFT	76.27	85.00
		Chirplet	90.56	87.50
4	GoogleNet + AlexNet	Poincare	76.94	72.50
		Stockwell	88.89	92.50
		STFT	70.62	90.00
		Chirplet	86.39	92.50
5	GoogleNet + ResNet18	Poincare	85.00	77.50
		Stockwell	88.06	85.00
		STFT	79.66	82.50
		Chirplet	92.78	85.00

TABLE 5. Performance Evaluation for the Best Performing Networks

S.No	Networks	Image	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	MCC
1	ResNet18	Chirplet	90.56	97.77	73.33	83.80	0.78
2	ResNet50	Chirplet	92.50	98.94	78.33	87.44	0.83
3	ShuffleNet + AlexNet	Stockwell	93.70	99.84	79.20	88.98	0.86
4	GoogleNet + ResNet18	Chirplet	92.78	98.96	80.00	87.67	0.84

Considering the number of computational parameters and the classification accuracy, different DL models are ensembled for identifying AF from various TFA techniques and their performances are represented in Table 4. Among the various networks, ShuffleNet, GoogleNet, and ResNet18 models have less number of parameters. GoogleNet is ensembled with ResNet18, AlexNet, and ShuffleNet, and reported the highest accuracy of 92.78% for Chirplet, 88.89%, 88.33% for Stockwell, respectively. AlexNet is ensembled with ShuffleNet and ResNet18 and recorded an accuracy of 93.70% and 90.56% for Stockwell and Chirplet, respectively. Among the various combinations, ShuffleNet ensembled with AlexNet provides the highest accuracy of 93.70% on the features obtained by ST. For identifying the best performing model, the other performances (such as precision, recall, F1-score, and MCC) are computed and represented in Table 5. ShuffleNet ensembled with AlexNet recorded precision, recall, F1-score,

TABLE 6. Comparison of Existing AF Classification and Detection Methods With Proposed Method

S.No	Author & Year	Techniques	Accuracy (%)
1	Parmar et al., (2023) [8]	DL	92.37
2	Zhang et al., (2023) [11]	ML	87.40
3	Anbalagan et al., (2024) [20]	ML	91.88
4	Proposed method	TFA-DL	93.70

and MCC of 99.84%, 79.20%, 88.98%, and 0.86, respectively. For PhysioNet CinC 2017 challenge database, GoogleNet ensemble with AlexNet achieved the maximum accuracy of 92.50% for both Stockwell and Chirplet, while in case of individual networks, GoogleNet and VGG16 achieved an accuracy of 90.00% for ST. The proposed method is compared with the existing AF detection algorithms and represented in Table 6.

IV. CONCLUSION

This manuscript proposes a DNN technique for AF classification using 2-D images generated from 1-D ECG signal, by various TFA methods (such as STFT, Chirplet-transform, ST, and Poincare plot), which provide informative features for AF diagnosis. This approach is trained and validated on PAF prediction challenge databases. From the experimental results it is concluded that ResNet18 attained the highest accuracy of 90.56% for the features obtained by Chirplet transform. Ensembling ShuffleNet and AlexNet achieved an impressive accuracy of 93.70% on Stockwell-transformed-images. Comparative experiments are carried out on PhysioNet CinC 2017 challenge database and the results unambiguously demonstrate the potential benefit and advantage of the proposed technique for AF classification. The developed model is intended to be used in clinics for identifying the severity of AF conditions effectively and helps reducing mortality over time due to arrhythmia. This method is robust to noise, which is addressed by augmentation, 1-D Gaussian smoothing by different kernels, and addition of Gaussian-white noise, etc., ECG morphology is not consistent among the patient, so in some cases CNN fails to accept the ECG with varied duration even they are cropped to same length. It results diagnostic information loss due to its less attention to the temporal variations. This method helps assisting Cardiologist in identifying other heart related diseases from ECG recordings in real-time.

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COMPLIANCE WITH ETHICAL STANDARDS

This manuscript satisfies and fulfills the compliance with ethical standards. There are no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data along with programs and supporting files will be made available on request. Few of the data are publicly available in the PhysioBank ATM.

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